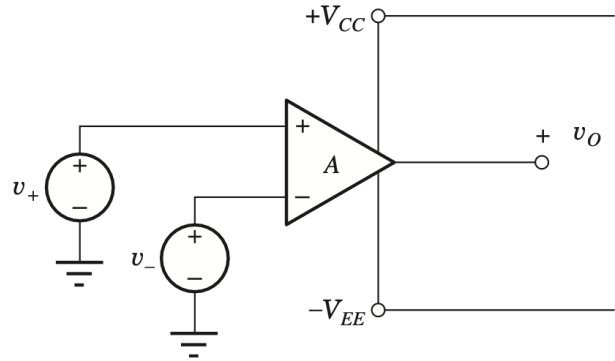


1) OP Amp

The basis to an OP Amp are the **differential components**

By defining the dc and ac components respectively as V_{IC} and $v_{i\pm}$, each current can be written as:

$$\begin{aligned}v_{I+}(t) &= V_{ic} + v_{i+}(t) \\v_{I-}(t) &= V_{ic} + v_{i-}(t)\end{aligned}$$



Differential Amplifier with Differential Input and Single Ended Output

We can now define:

$$\text{Differential Component: } v_{id}(t) = v_{i+}(t) - v_{i-}(t)$$

$$\text{Common Component: } v_{ic}(t) = \frac{v_{i+}(t) + v_{i-}(t)}{2}$$

Then the following relationships hold

$$\begin{aligned}v_{i+}(t) &= v_{ic}(t) + v_{id}(t)/2 \\v_{i-}(t) &= v_{ic}(t) - v_{id}(t)/2\end{aligned}$$

OP Amp

An **OP Amp** is defined by the following variables:

Name	Formula	Ideal Value	Real value
Differential Voltage Gain	$A_{dm} = \frac{v_o}{v_{id}} \mid v_{ic}=0$	∞	60 – 120dB
Common Voltage Gain	$A_{cm} = \frac{v_o}{v_{ic}} \mid v_{id}=0$	0	
	$\text{CMRR} = \left \frac{A_{dm}}{A_{cm}} \right $	∞	80 – 120dB
Input Resistance	R_{id}, R_{ic}	∞	1 – 100MΩ
Output Resistance	R_o	0	10 – 100Ω

Under the **linearity assumption** we have that the output voltage can be written as:

$$v_o(t) = A_{dm}v_{id}(t) + A_{cm}v_{ic}(t)$$

The supply voltage imposes some natural limits to the amplifier, by assuming that $V_{SS} = -V_{DD}$:

$$V_{ss} < V_{o,min} < v_o < V_{o,max} < V_{dd}$$

At the **dc operating point** it holds that the voltage difference between $V_I = V_{IC} = V_o = 0$, and thus $v_o(t) = A_{dm}v_{id}$, this implies:

$$\frac{V_{SS}}{A_{dm}} \leq v_{id} \leq \frac{V_{DD}}{A_{dm}}$$

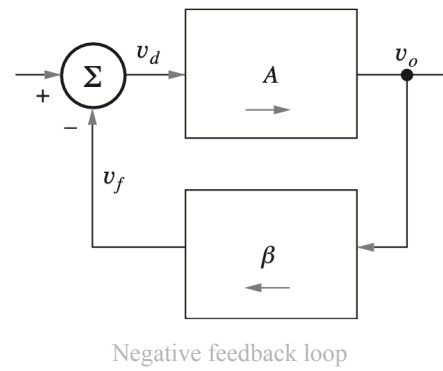
In order for the amp to operate in the linear region we need a **negative feedback**

First we write some relations we can easily deduce from the diagram:

$$\begin{cases} v_0 = A \cdot v_d \\ v_d = v_s - v_f \\ v_f = \beta \cdot v_0 \end{cases} \implies v_0 = \frac{A}{1 + \beta A} v_s \xrightarrow{A \rightarrow \infty} \frac{1}{\beta} v_s$$

from here we can also find v_d :

$$v_d = v_s - v_f = v_s - \beta v_0 = \frac{1}{1 + \beta A} v_s \xrightarrow{A \rightarrow \infty} 0$$



Moreover we have a **virtual short circuit** that states that $v_+ = v_-$ and that no current is entering the op-amp.

Various Types of OP-Amp configurations

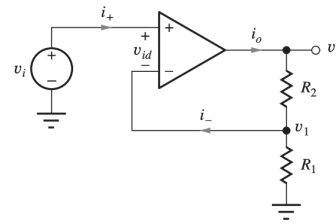
NON INVERTING AMPLIFIER

A non inverting amplifier has the following values:

$$A_v = 1 + \frac{R_2}{R_1}$$

$$R_i \rightarrow \infty$$

$$R_o = 0$$



Circuit

Proofs:

Voltage Gain:

By the vsc (virtual short-circuit) we have that: $v_i = v_+ = v_- = v_1$ and by the voltage divider:

$$v_1 = \frac{R_1}{R_1 + R_2} v_0 = \frac{1}{1 + \frac{R_2}{R_1}} v_0 \implies A_v = \frac{v_0}{v_i} = \frac{v_0}{\frac{1}{1 + \frac{R_2}{R_1}} v_0} = 1 + \frac{R_2}{R_1}$$

R_i :

We know that $R_i = v_i / i_i$ but since $i_i = 0$ due to the vsc, we have that R_i diverges

R_o :

We know that $R_o = i_o / v_o$ in this case $i_o = 0$ and thus the resistance is null

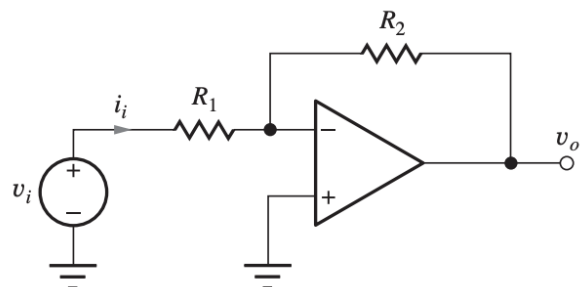
INVERTING AMPLIFIER

An inverting amplifier has the following values:

$$A_{v0} = -\frac{R_2}{R_1}$$

$$R_i = R_1$$

$$R_o = 0$$



Proof:

Voltage Gain:

Due to the vsc we know $v_- = v_+ = 0$ and that $i_1 = i_2 = i$. Then we know that $i = v_i/R_1$ and that $v_o = -R_2 i = -\frac{R_2}{R_1} v_i$ $\square$

R_i :

We know that $R_i = v_i/i_i$. In this case $i_i = i = v_i/R_1 \implies R_i = R_1$.

R_o :

We know that $R_o = v_o/i_o$ but since $v_i = 0$ to measure R_o then also $i_o = i = 0$.

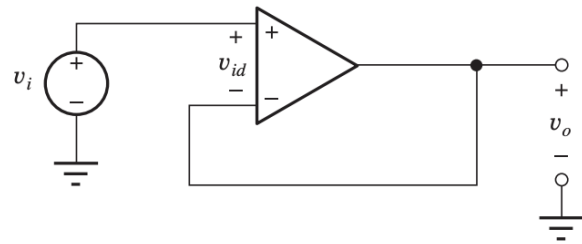
VOLTAGE BUFFER

An inverting amplifier has the following values:

$$A_{v0} = 1$$

$$R_i \rightarrow \infty$$

$$R_o = 0$$

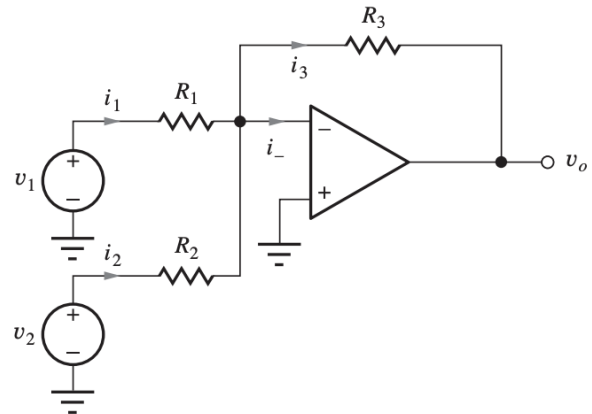


Circuit

SUMMING AMPLIFIER

To get a generalized form suppose we have n resistances at the load and the resistance at negative feedback is R_F :

$$v_o = -R_F \sum_{k=1}^n \frac{v_k}{R_k}$$



Circuit

Proof:

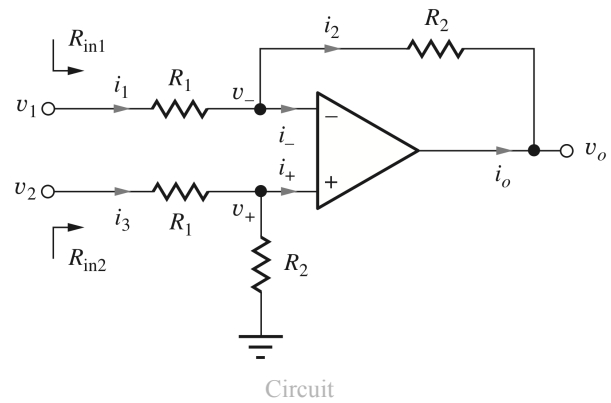
Notice that $v_+ = v_i = 0$ then $i_k = \frac{v_k}{R_k} \quad \forall \quad \text{voltage source. Clearly } i_F = \sum_{k=1}^n i_k, \text{ then}$

$$v_o = -R_F i_F = -R_F \sum_{k=1}^n i_k = -R_F \sum_{k=1}^n \frac{v_k}{R_k} = -\sum_{k=1}^n R_F R_k v_k.$$

DIFFERENCE AMPLIFIER

$$v_0 = -\frac{R_2}{R_1}(v_1 - v_2)$$

This can be seen as a superposition of an inverting and a non inverting op-amp



It is possible to find circuits with transient components, but the analysis is the same when operating in Laplace domain and by considering the "transfer functions", and thus the impedances and a *bode plot* representation to better analyze the response.

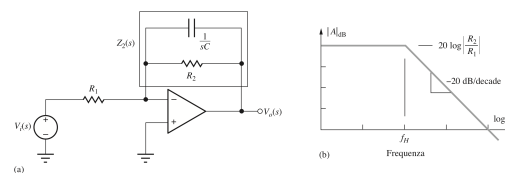
For the already seen Op-Amps nothing changes in Laplace domain.

First, a quick recap on the response of the basic circuit components:

R	$\xrightarrow{\mathcal{L}}$	R	$\xrightarrow{\text{ac}}$	R
C	$\xrightarrow{\mathcal{L}}$	$\frac{1}{sC}$	$\xrightarrow{\text{ac}}$	$\frac{1}{j\omega C}$
L	$\xrightarrow{\mathcal{L}}$	sL	$\xrightarrow{\text{ac}}$	$j\omega L$

LOW-PASS FILTER/AMPLIFIER

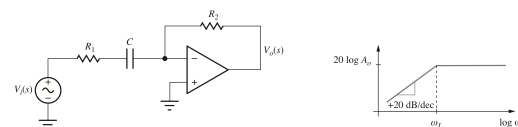
$$\begin{aligned} Z_1 &= R_1 \\ Z_2 &= \frac{R_2}{1+sR_2C} \\ H(s) &= -\frac{Z_2}{Z_1} = -\frac{R_2}{R_1} \frac{1}{1+sR_2C} \end{aligned}$$



Circuit+Bode Plot

HIGH PASS FILTER/AMPLIFIER

$$\begin{aligned} Z_1 &= R_1 + \frac{1}{sC} = \frac{sCR_1+1}{sC} \\ Z_2 &= R_2 \\ H(s) &= -\frac{Z_2}{Z_1} = -\frac{sR_2C}{1+sR_1C} \end{aligned}$$

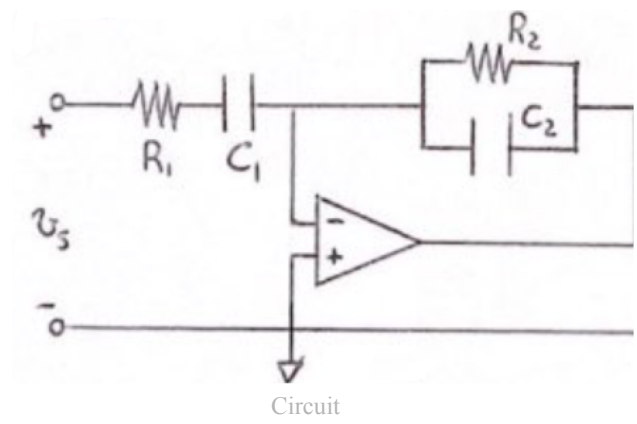


Circuit+Bode Plot

BAND PASS FILTER/AMPLIFIER

This is a mix between a low and a high pass filter

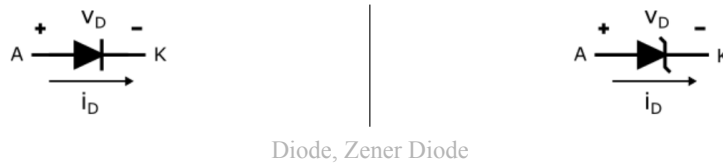
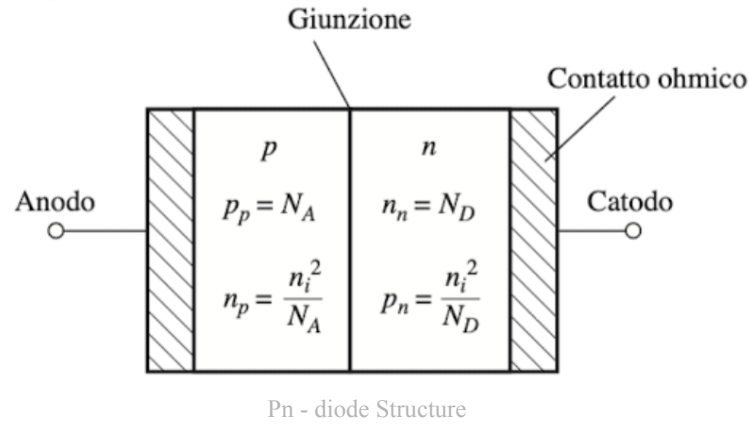
$$\begin{aligned} Z_1 &= R_1 + \frac{1}{sC_1} = \frac{sC_1R_1+1}{sC_1} \\ Z_2 &= \frac{R_2}{1+sR_2C_2} \\ H(s) &= -\frac{Z_2}{Z_1} = -\frac{sR_2C_1}{(1+sR_1C_1)(sR_2C_2)} \end{aligned}$$



Recall that when I transform to the Laplace domain I must ignore any constant input and set it to 0 (ground) since it doesn't depend on time.

2) Diode

This will Just be a quick recap of the PWL model approximation.

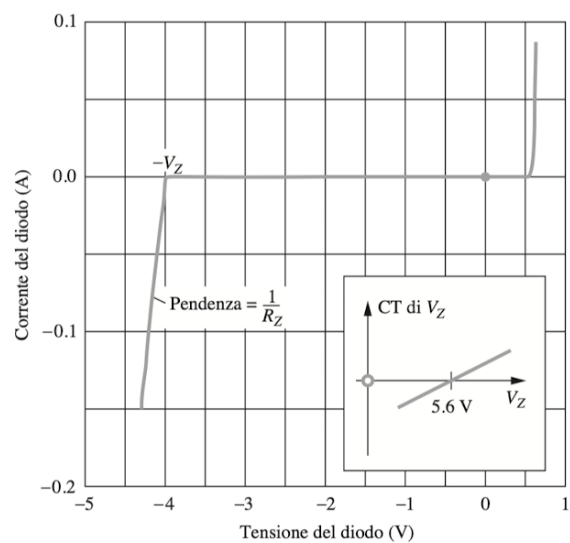


As seen in the graph in the image, it is possible to identify three different operation regions for the diode. From left to right:

- Breakdown Region** $\in [-\infty, -V_Z]$
- Reverse Region** $\in [V_Z, V_{ON}]$
- Forward Region** $\in [V_{ON}, +\infty]$

In our approximation we will set:

$$i_D = \begin{cases} \frac{V_D + V_Z}{R_Z}, & V_D < -V_Z \\ 0 & -V_Z < V_D < V_{ON} \\ \frac{V_D - V_{ON}}{R_{ON}}, & V_D > V_{ON} \end{cases}$$



If $R_{ON} = 0$ then the PWL model becomes:

$$\begin{cases} i_D = \frac{V_D + V_Z}{R_Z}, & V_D < -V_Z & \text{i.e. break down region} \\ i_D = 0 & -V_Z < V_D < V_{ON} & \text{i.e. reverse region} \\ V_D = V_{ON}, & V_D > V_{ON} & \text{i.e. forward region} \end{cases}$$

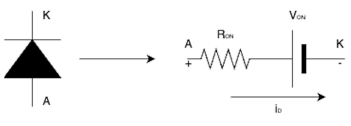
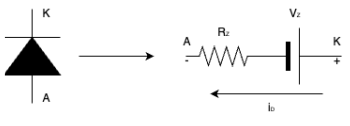
This condition is often assumed.

Recall that $V_D = V_A - V_K$

Remark (Breakdown Region and Zener Diode).

The diode must **avoid the breakdown region unless it is a zener diode**. In the other 2 regions they behave exactly the same.

On an analysis of the circuit we can assume one of the three operation regions and solve the circuit using an equivalent model instead of the diode. The solution is valid if the hypothesis holds with the results:

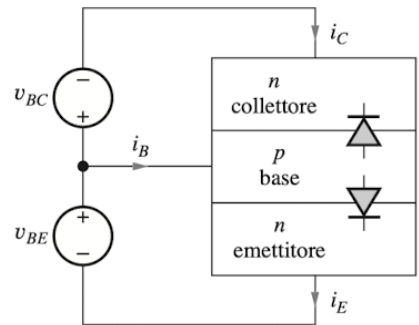
$R_{ON} = 0$	Forward	Reverse	Breakdown
Condition	$i_D > 0$	$-V_Z < V_D < V_{ON}$	$V_D < V_Z, -I_{Z_{max}} < i_D < I_{Z_{max}}$, must be zener
Circuit	Replace diode with voltage source $V_D = V_{ON}$	Replace diode with open circuit $i_D = 0$	Replace diode with voltage source $V_D = -V_Z$
$R_{ON} \neq 0$	Forward	Reverse	Breakdown
Condition	$V_D > V_{ON}, i_D > 0$	$-V_Z < V_D < V_{ON}$	$V_D < V_Z, -I_{Z_{max}} < i_D < I_{Z_{max}}$, must be zener
Circuit	$V_D = V_{ON}$  Forward Region\	Replace diode with open circuit $i_D = 0$	$V_D = -V_Z$  Breakdown Region\

Remark (Current Direction).

It is clear that in a diode the current can only pass in the direction where the triangle is "pointing". Therefore in a circuit analysis it can be used to make some informed guess on the behaviour of the current.
In a Zener diode the current can also go the opposite way, so it is good to start with that hypothesis

3) Bipolar Junction Transistor (BJT)

As it is shown in the image, the BJT is made from 2 diodes.



PNP-BJT Structure



NPN VS PNP

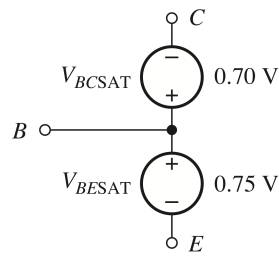
We can identify 4 operation regions (NPN BJT used for table):

BE Junction	BC Junction	BJT Operation Region
forward	forward	saturation (used in digital circuits)
reverse	reverse	cut-off (used in digital circuits)
forward	reverse	forward-active (active) (used in analog circuits)
reverse	forward	reverse active (usually not used)

Forward Active Region Approx. Model:

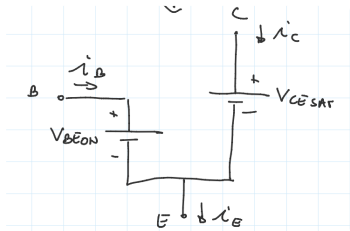
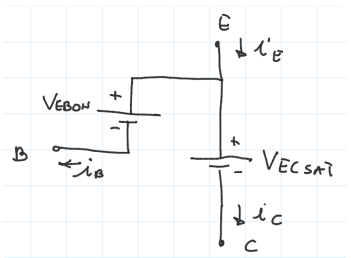
Type	NPN	PNP
Condition	if $i_B > 0$ and $V_{BC} < V_{BC_{ON}} \Rightarrow V_{CE} > V_{CE_{sat}}$	if $i_B > 0$ and $V_{CB} < V_{CB_{ON}} \Rightarrow V_{EC} > V_{EC_{sat}}$
Result	$V_{BE} = V_{BE_{ON}}$ and $i_C = \beta \cdot i_B$, $i_E = (\beta + 1)i_B$	$V_{EB} = V_{EB_{ON}}$ and $i_C = \beta \cdot i_B$, $i_E = (\beta + 1)i_B$
Approx. Model	(b) NPN BJT Active Approx. Model\ The diagram shows a simplified circuit model for an NPN BJT in the forward active region. It includes a 0.7 V DC voltage source representing the base-emitter junction, with the positive terminal at the base (B) and the negative terminal at the emitter (E). The base current i_B flows into the base terminal. A dependent current source is connected between the collector (C) and emitter (E) terminals, with its value given by $i_C = \beta i_B$. The collector current i_C flows out of the collector terminal, and the emitter current i_E flows out of the emitter terminal.	

Saturation Region Approx. Model

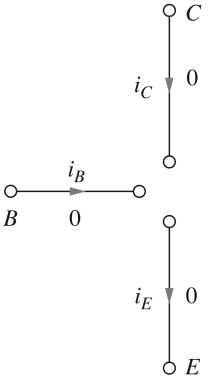


NPN BJT Saturation Region Approx. Model

By setting $V_{CE} = V_{BE_{ON}} - V_{BC_{ON}} = V_{CE_{sat}}$ we get to the following table:

Type	NPN	PNP
Condition	if $i_B > 0$ and $i_C < \beta_F i_B$	if $i_B > 0$ and $i_C < \beta_F i_B$
Result	$V_{BE} = V_{BE_{sat}} = V_{BE_{ON}}, V_{CE} = V_{CE_{sat}}$	$V_{EB} = V_{EB_{sat}} = V_{EB_{ON}}, V_{EC} = V_{EC_{sat}}$
Approx. Model	 Even More Approx.\	 Even More Approx.\

Cut-Off Region Approx. Model

Type	NPN	PNP
Condition	if $V_{BE} < V_{BE_{ON}}$ and $V_{BC} < V_{BC_{ON}}$	if $V_{EB} < V_{EB_{ON}}$ and $V_{CB} < V_{CB_{ON}}$
Result	all currents = 0	all currents = 0
Approx. Model	 Cut-Off Region Approx. Model\	

The **dissipated power** is given by:

$$P_{BJT} = I_C V_{CE} + I_B V_{BE}$$

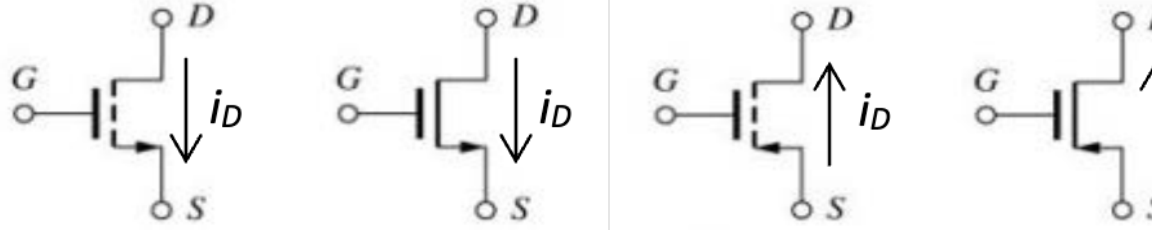
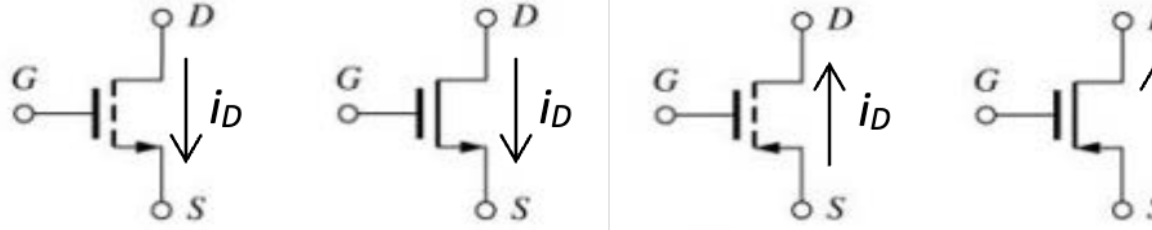
4) Metal Oxide Semiconductor Field Effect Transistor (MOSFET)

The MOSFET is described by 4 parameters:

- V_{TH} : threshold voltage
- K'_n : normalized transconductance factor
- W/L : geometric form factor
- λ : channel length modulator factor

Moreover we have that: $K_n = K'_n \frac{W}{L} = \mu C_{ox} \frac{W}{L}$

notice that μ is either μ_n or μ_p if it's a n or p type mosfet

	nMosfet	pMosfet
Symbol		
	nMosfet\	pMosfet\

Saturation Region:

	nMosfet	pMosfet
Conditions	if $V_{GS} > V_{TH}$ and $V_{DS} > V_{OV} = V_{GS} - V_{TH}$	if $V_{GS} < \ V_{TH}\ $ and $V_{DS} < V_{OV} = V_{GS} - \ V_{TH}\ $
Result	$i_D = \frac{K_n}{2} (V_{GS} - V_{TH})^2$	$i_D = \frac{K_p}{2} (V_{GS} - V_{TH})^2$

Triode Region:

	nMosfet	pMosfet
Conditions	if $V_{GS} > V_{TH}$ and $V_{DS} < V_{OV} = V_{GS} - V_{TH}$	if $V_{GS} < V_{TH}$ and $V_{DS} > V_{OV} = V_{GS} - V_{TH}$
Result	$i_D = K_n \left((V_{GS} - V_{TH})V_{DS} - \frac{V_{DS}^2}{2} \right)$	$i_D = K_p \left((V_{GS} - V_{TH})V_{DS} - \frac{V_{DS}^2}{2} \right)$

Cut-Off Region:

	nMosfet	pMosfet
Conditions	if $V_{GS} < V_{TH}$	if $V_{GS} > V_{TH}$
Result	$i_D = 0$	$i_D = 0$

5) Bode Diagram

The **Bode diagram** is a useful tool to represent a frequency response in a graph using a pwl (piecewise linear) model that asymptotically approximates the real plot. There are two types of plots, the **amplitude** and the **phase** plot, both in logarithmic scale.

Why log scale?

We know that $W(jw)$ can be represented in trigonometric form as $W(jw) = |W(jw)|e^{j \arg(W(jw))}$. The, using the neperian complex logarithmic we end up with an expression of form $a + jb$:

$$\ln(W(jw)) = \ln(|W(jw)|) + \ln(e^{j \arg(W(jw))}) = \ln(|W(jw)|) + j \cdot \arg(W(jw))$$

Moreover we use the **bilogarithmic scale**, that is: $20 \log_{10}()$

Recall how a transfer function is usually expressed:

$$W(s) = \frac{n(s)}{d(s)} = \frac{\sum_{m=0}^{\deg(n(s))} b_m s^m}{\sum_{n=0}^{\deg(d(s))} a_n s^n}$$

which can be rewritten as

$$W(s) = \frac{b_m}{a_n} \frac{(s - z_1) \dots (s - z_m)}{(s - p_1) \dots (s - p_2)}$$

With some more algebraic manipulation, and by considering the *Fourier transform* (i.e. the phase representation) it is possible to show that **constant terms, zeroes, poles (real and complex)** give different types of contribution to the bode plot, but all these **contributions are additive**

Remark (Electronics).

In electronics we will have that $\frac{b_m}{a_n} = (\frac{\omega_0}{s})^r$ which includes the r poles in zero and has $(\omega_0)^r = (\frac{b_m}{a_n})^r$

From now on $\log_{10}$ will be simply referred as $\log$

Remark (Semilog scale vs log scale).

Wolfram alpha, and many other representations are in **semilogarithmic scale**. The x axis is in w , not $\log w$.

Notice that in log scale the slope becomes $[dB/\text{decade}^2] \xrightarrow{\text{log scale}} [dB/\text{decade}]$.

I will write all results in semilog scale. However the plots are in semilogarithmic scale in order to show more of the graph

5.2) Constant Term

The first term (K_B) is called **Evans gain** and it is important since it is constant, and it also **determines the gain/attenuation**. A simple way to find it is by finding the **DC/Static gain**:

Definition 1 (DC/Static Gain).

Given a transfer function $W(s)$ with no pole in the origin, we define its gain as the value in $W(0) = K_B$. Then we can group this term in the expression.

This can also be extended to the case with some poles in the origin, by just ignoring the term $\frac{w_0}{s}$ in the calculation

Suppose we have

$$W(s) = K_B$$

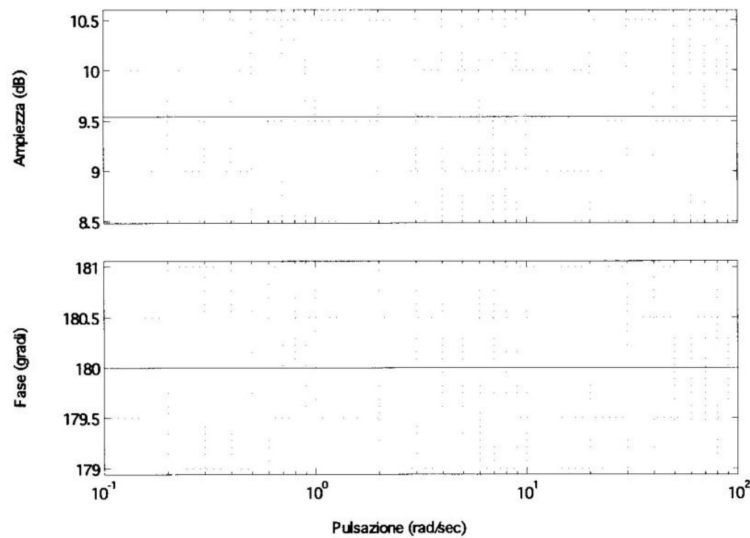
AMPLITUDE PLOT

It is just a straight line of form $y = 20 \log |K_B|$

If $|K_B| > 1$ we have a gain in the output, otherwise we have an attenuation

PHASE PLOT

Since $K_B \in \mathbb{R}$ we have that if $K_B > 0$ the phase is null, otherwise it is $\pm 180^\circ$ it doesn't matter which one



Bode diagram of $W(s)=-3$

5.3) Pole in Origin

Suppose we have

$$W(s) = \frac{1}{s^v}$$

where v is the degree which coincides with the amount of poles in the origin.

AMPLITUDE PLOT

The transfer function will be $|W(jw)|_{dB} = -20v \log w$

This means that the resulting plot will be a line passing through the origin $[0, 0]$ and it will:

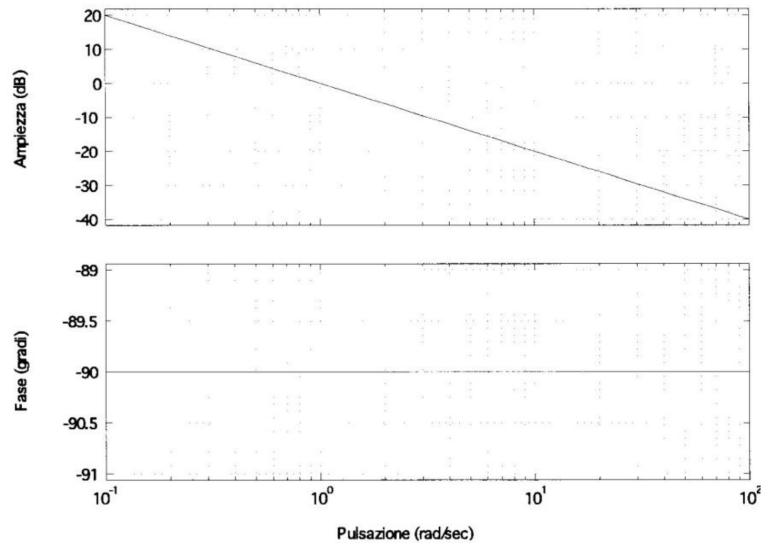
- $v > 0$ go downwards with slope $-20|v| [dB/decade]$
- $v < 0$ go upwards with slope $20|v| [dB/decade]$

PHASE PLOT

The transfer function will be $\arg(W(j\omega)) = -v \cdot 90^\circ$.

This means that the resulting line will be of form $y = \mp 90^\circ$

- $v > 0 \implies -|v|90^\circ$
- $v < 0 \implies |v|90^\circ$



Bode plot of $W(s)=1/s$

Remark.

A zero in the origin is the equivalent of a pole in the origin, just with different exponential term

Corollary 2 (Pole in Zero with Gain).

As explained before in electronics we will have $(\frac{\omega_0}{s})^r$ as the first term which includes the r poles in zero and has $(\omega_0)^r = (K_B)^r$.

*This is actually useful since you can draw just **one line with the contribution of the zero and constant term included**.*

If we have poles in the origin ($r > 0$) this line has negative slope ($-20|r|$ [dB/decade]) and will pass through $[|r| \log \omega_0, 0]$.

Otherwise, if there are zeros in the origin ($r < 0$) the line has positive slope ($20|r|$ [dB/decade]) and will still pass through $[|r| \log \omega_0, 0]$.

Notice that the lines change sign in the slope and it would suggest that the intersection points would also be switched, but ω_0 becomes its reciprocal with $r < 0$ and thus the effects cancel out.

The phase plot depends on the sign of $(\omega_0)^r$.

If it is positive it will give zero contribution to the phase and thus the phase plot will be identical to the case of a pole in the origin.

Otherwise, the phase will be shifted by a factor of 180° .

5.4) Real Poles and Zeroes

Suppose we have

$$W(s) = (1 + sT)^\mu$$

This is valid for zeroes if $\mu > 0$, otherwise it is valid for poles, and $|\mu|$ is the number of poles/zeroes. Both cases are centered in $-1/T$, then we end up with:

$$\begin{aligned}\log |W(jw)| &= \mu \log \sqrt{1 + (wT)^2} \\ \arg(W(jw)) &= \mu \arctan(wT)\end{aligned}$$

Definition 3 (Cutoff frequency).

We define the **cutoff frequency** as

$$w_c = \frac{1}{|T|}$$

This frequency is the minimum required frequency to not have a 0 in the Bode diagram approximation

We can rewrite all the poles and zeroes as $s - p \rightarrow 1 + \frac{s}{p} \rightarrow 1 + Ts = 1 + \frac{s}{w_c}$

I will write the results according to w_c

Here we see the first implementation of the pwl model, since we will consider the following 2 approximations:
 $|wT| \ll 1, |wT| \gg 1 \implies w \ll 1/|T|, w \gg 1/|T|$.

Then we have the two cases:

- $w \ll w_c$

$$\begin{aligned}\log |W(jw)| &\approx \mu \log 1 = 0 \\ \arg(W(jw)) &\approx \mu \arg 1 = 0^\circ\end{aligned}$$

- $w \gg w_c$

$$\begin{aligned}\log |W(jw)| &\approx \mu \log wT = \mu(\log w + \log T) \\ \arg(W(jw)) &\approx \mu \arg 1 = \mu \cdot \text{sign}(T) \cdot 90^\circ\end{aligned}$$

From here it is clear that **real zeroes and poles behave in the same way, but with inverted sign.**

AMPLITUDE PLOT

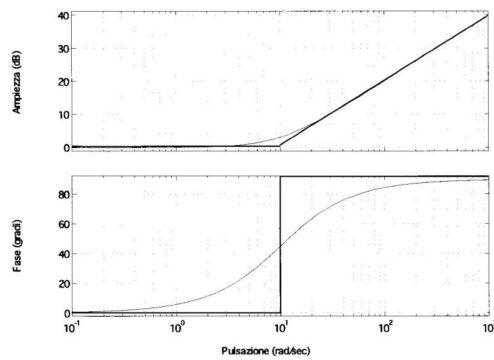
It will be $y = 0$ until point $[\log w_c, 0]$ (cutoff frequency) on the x-axis, then it will be a line going upwards (zero) or downwards (pole) with slope $\pm 20|\mu| [dB/\text{decade}]$.

Remark (Zero in origin).

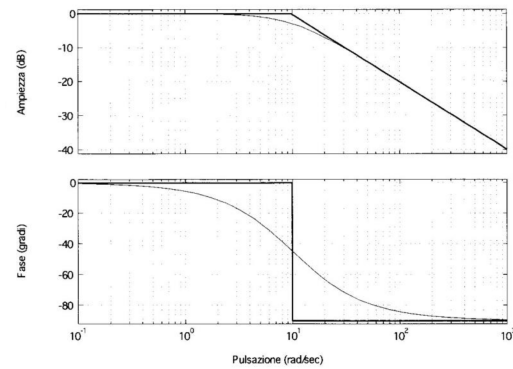
From here we can see that usually the slope goes upwards for zeroes, and downwards for poles. You can also apply this reasoning for Zeroes and Poles in the origin.

PHASE PLOT

The phase will be 0 until point $[\log(w_c) - 1, 0]$, then it will be a straight line connecting to the point $[\log(w_c) + 1, \pm |\mu| \cdot 90^\circ]$ if it is zero or pole.



Bode diagram of $W(s)=1+0.1s$



Bode diagram of $W(s)=(1+0.1s)^{-1}$

The phase plot has a wrong approximation, but from the real curve it is evident how the slope should be

5.5) Complex Conjugate Poles and Zeroes

Suppose we have

$$W(s) = \left(1 + 2\epsilon \frac{s}{w_n} + \left(\frac{s}{w_n} \right)^2 \right)^\mu$$

Now we have:

$$|W(jw)|_{dB} = 10\mu \log \left[\left(1 - \left(\frac{w}{w_n} \right)^2 \right)^2 + \left(2\epsilon \frac{w}{w_n} \right)^2 \right]$$

$$\arg(W(jw)) = \begin{cases} \arctan \left(\frac{2\epsilon \frac{w}{w_n}}{1 - \frac{w^2}{w_n^2}} \right), & w < w_n \\ \text{sign}(\epsilon) \cdot \frac{\pi}{2}, & w = w_n \\ \arctan \left(\frac{2\epsilon \frac{w}{w_n}}{1 - \frac{w^2}{w_n^2}} \right) + \text{sign}(\epsilon) \cdot \pi, & w > w_n \end{cases}$$

Now we can find the asymptotic approximations for $w \ll w_n$, $w \gg w_n$:

- $w \ll w_n$

$$\log |W(jw)| \approx \mu \log 1 = 0$$

$$\arg(W(jw)) = \mu \arg 1 = 0^\circ$$

- $w \gg w_n$

$$\log |W(jw)| \approx 2\mu \log \left(\frac{w}{w_n} \right) = 2\mu (\log w - \log w_n)$$

$$\arg(W(jw)) = \mu \cdot \text{sign}(\epsilon) \cdot \pi = \mu \cdot \text{sign}(\epsilon) \cdot 180^\circ$$

So we have find another case of **cutoff**.

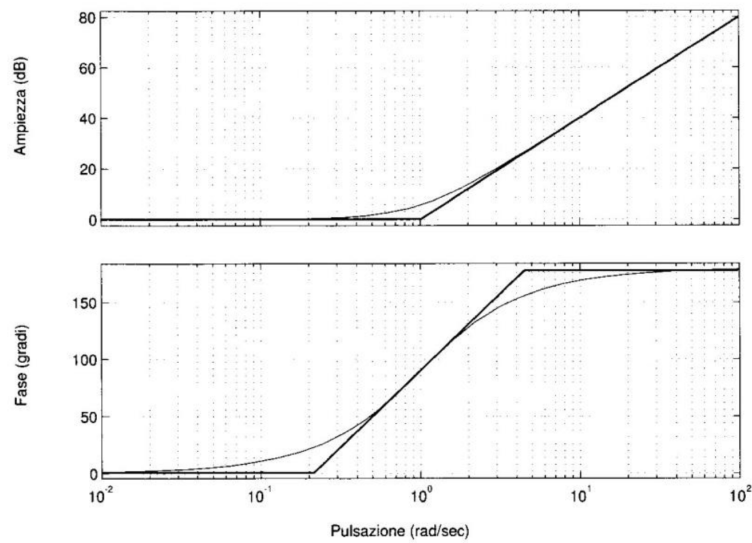
Again we have that μ shows the number of poles or zeroes

AMPLITUDE PLOT

It will be zero until $[\log w_n, 0]$, then it will be a line going upwards (zero) or downwards (pole) with slope $\pm 40|\mu|[\text{dB/decade}]$.

PHASE PLOT

The phase will be 0 until point $[\log(w_n) - 1, 0]$, then it will be a straight line connecting to the point $[\log(w_n) + 1, \pm |\mu| \cdot 180^\circ]$ if it is zero or pole.



Bode plot of $W(s)=s^2+1.9s+1$

Here the approximation of the phase is correct. Moreover in the example we have $w_n = 1$, $\epsilon = 0.95$, $\mu = 1$

5.6) How to Draw a Bode Diagram With Multiples Zeroes and Poles

It's actually pretty easy by recalling the **additive** property that was given at the start

AMPLITUDE PLOT

1. Draw every single line corresponding to every zero/pole/constant term
2. In the resulting plot, every point will have a slope corresponding to the sum of the slopes of every line in that point
3. add the constant value gain/attenuation

Example.

This is a very dumb example, but consider

$$W(s) = \frac{1-s}{1-s}$$

This is clearly 1. And the bode plot will be a straight line of form $y = 0$. But suppose we are a dumb student who doesn't notice this and applies every rule.

1. Zero in 1, that means it gives 0 contribution until $[1, 0]$ and then rises with slope $+20[dB/decade^2]$
2. Pole in 1, that is 0 contribution until $[1, 0]$ and then goes downwards with slope $-20[dB/decade^2]$
3. Until $[1, 0]$ there is no contribution, then after that point the 2 contributions are $\pm 20[dB/decade^2]$ and thus $20 - 20 = 0[dB/decade^2]$

Applying the rules gives the same result of the "direct" computation $\square$

PHASE PLOT

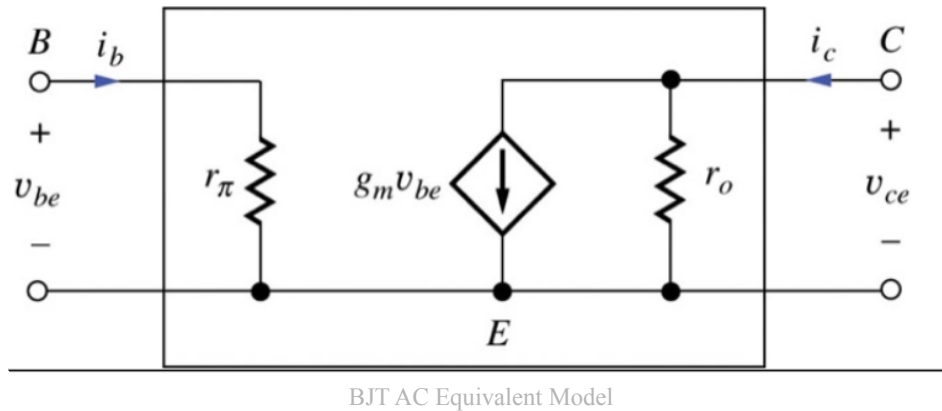
1. Draw every single line corresponding to every zero/pole/constant term
2. In the resulting plot, every point will have a slope corresponding to the sum of the slopes of every line in that point
3. When the phase has a constant attribution add\subtract it

6) Small Signal Regime Analysis

To give a quick idea of what this analysis represents imagine solving a circuit in dc. You find a result of how it functions (voltages, currents, operating regions) but you don't know how it handles information. The SSR is an oscillating voltage applied on top of ur DC signal and this is the actual transmitter of information. To solve a circuit we will first find how it operates (DC) and then, via some equivalent models, we will find how it behaves with AC current. We will see that transistors behave as amplifiers.

6.1) BJT Equivalent Models

PNP and NPN BJT's have the same equivalent model



The three important parameters are:

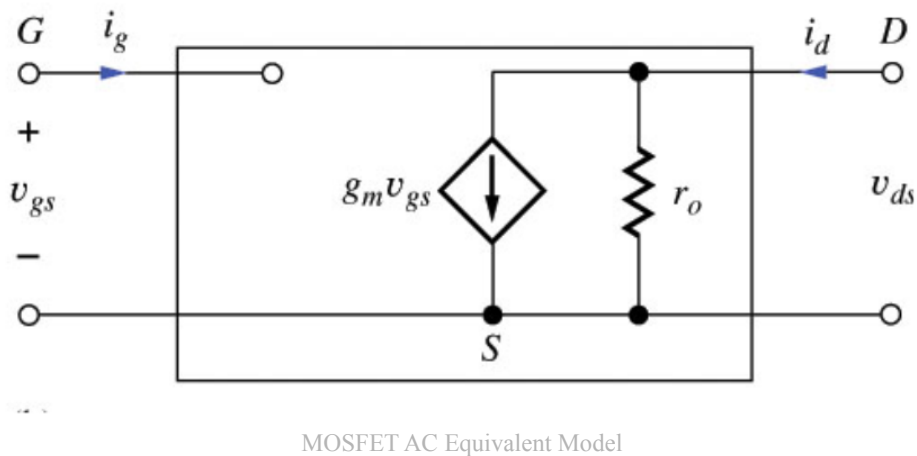
$$g_m = \frac{I_C}{U_T} \approx 40 \cdot I_C$$

$$r_\pi = \frac{\beta}{g_m}$$

$$r_o = \frac{V_A + |V_{CE}|}{I_C} \approx \frac{V_A}{I_C}$$

6.2) MOSFET Equivalent Models

n and p- MOSFET have the same equivalent model *but with slightly different equations.*

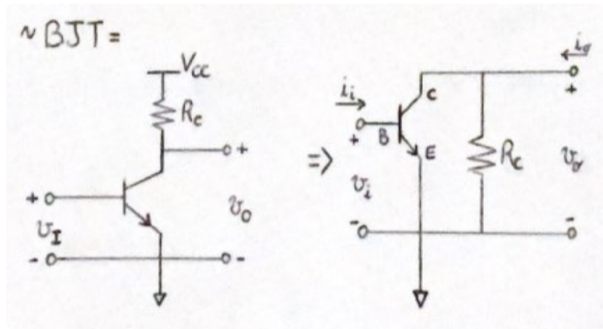


nMOSFET	pMOSFET
$g_m = \frac{2I_D}{V_{GS} - V_{TH}} = \sqrt{2K_n I_D} = K_n(V_{GS} - V_{TH})$	$g_m = \frac{2I_D}{\ V_{GS} - V_{TH}\ } = \sqrt{2K_n I_D} = K_n\ V_{GS} - V_{TH}\ $
$r_o = \frac{1}{g_m} \frac{1 + \lambda V_{ds}}{\lambda I_D} \approx \frac{1}{\lambda I_D}$	$r_o = \frac{1}{g_m} \frac{1 + \lambda \ V_{ds}\ }{\lambda I_D} \approx \frac{1}{\lambda I_D}$

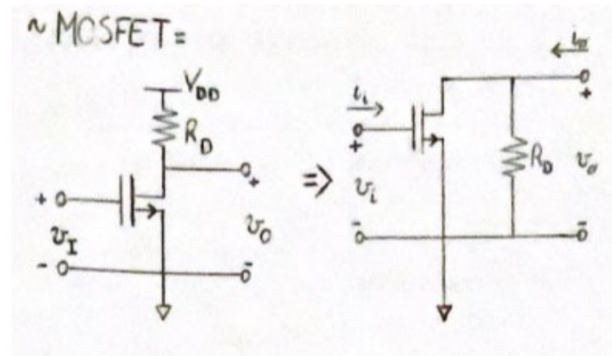
6.3) Integrated Circuits

This is the *juicy part*. Usually we can find more complex circuits that happen to behave in a way that can be approximated again by an amplifier. The list is quite long but it's quite repetitive in the end, just practice until you get a feel on how to solve the exercises.

6.3.1) Common Emitter, Common Source (CE, CS)

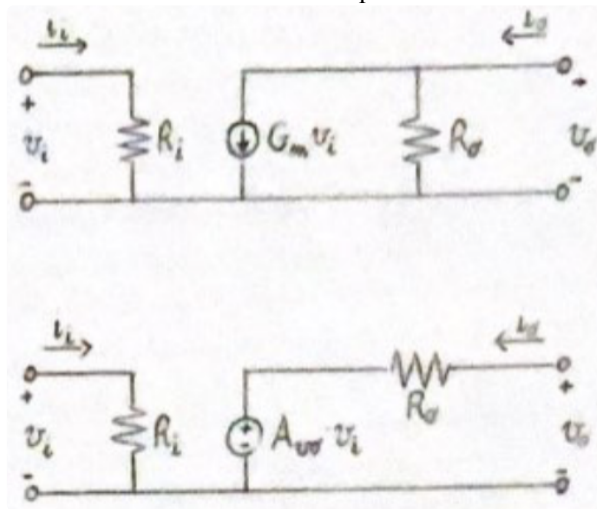


BJT CE



MOSFET CE

The equivalent model is the same for both but with different equations:

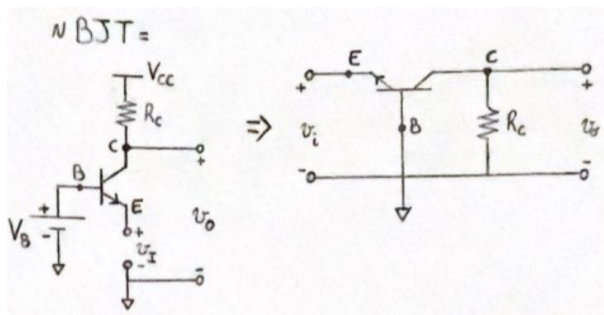


CE Equivalent Model

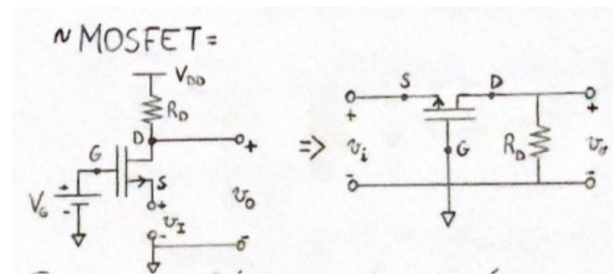
	BJT	MOSFET
R_i	$r_\pi = \frac{\beta}{I_C} = \frac{\beta U_T}{I_C} = \frac{v_i}{i_i}$	∞
G_m	$g_m = \frac{I_C}{U_T} \approx 40 I_D$	$g_m = \frac{2I_D}{V_{OV}} = K_n V_{OV}$
R_o	$r_o // R_C = \frac{v_o}{i_o} \parallel v_i$ with $r_o = \frac{V_A + \ V_{CE}\ }{I_C} \approx \frac{V_A}{I_C}$	$r_o // R_D$ with $r_o = \frac{1}{\lambda I_D}$
A_{v0}	$-G_m R_o$	$-G_m R_o$
i_o	$G_m v_i + \frac{v_o}{R_o}$	

6.3.2) Common Emitter, Common Drain with Degenerative Resistance (CER, CSR)

6.3.3) Common Base, Common Gate (CB, CG)

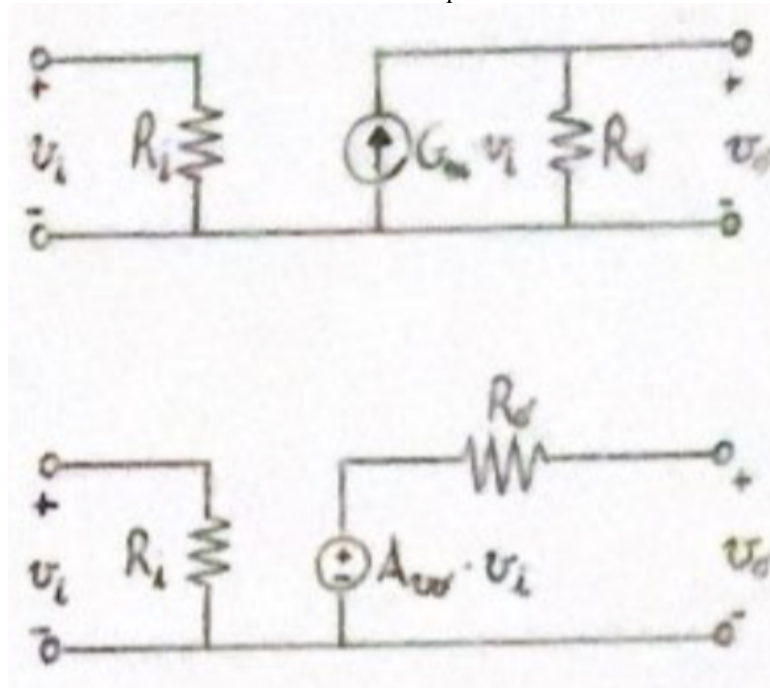


BJT CB



MOSFET CB

The equivalent model is the same for both but with different equations:



CE Equivalent Model

This works under the **assumption** that $r_o \gg R_C$, $r_o \gg R_E$.

	BJT	MOSFET

6.3.4) Common Collector, Common Drain (CC, CD)

6.3.5) Current Mirror

6.3.6) Differential Couple

6.3.7) Common Mode